vertiq

VERTIQ EXPLANATION

Vertiq, explained simply

A self-improving decision intelligence system: what it does, how it works, an example, the questions it answers in each industry, and why a database query is not enough.

Predicts

what is likely to happen next to a customer, a machine, a patient or a shipment

Measures

whether an action actually changed the outcome, or would have happened anyway

Improves

itself, but only replaces a version after proving the new one is better

WHAT VERTIQ IS

A system that learns from what really happened, and proves itself before you rely on it

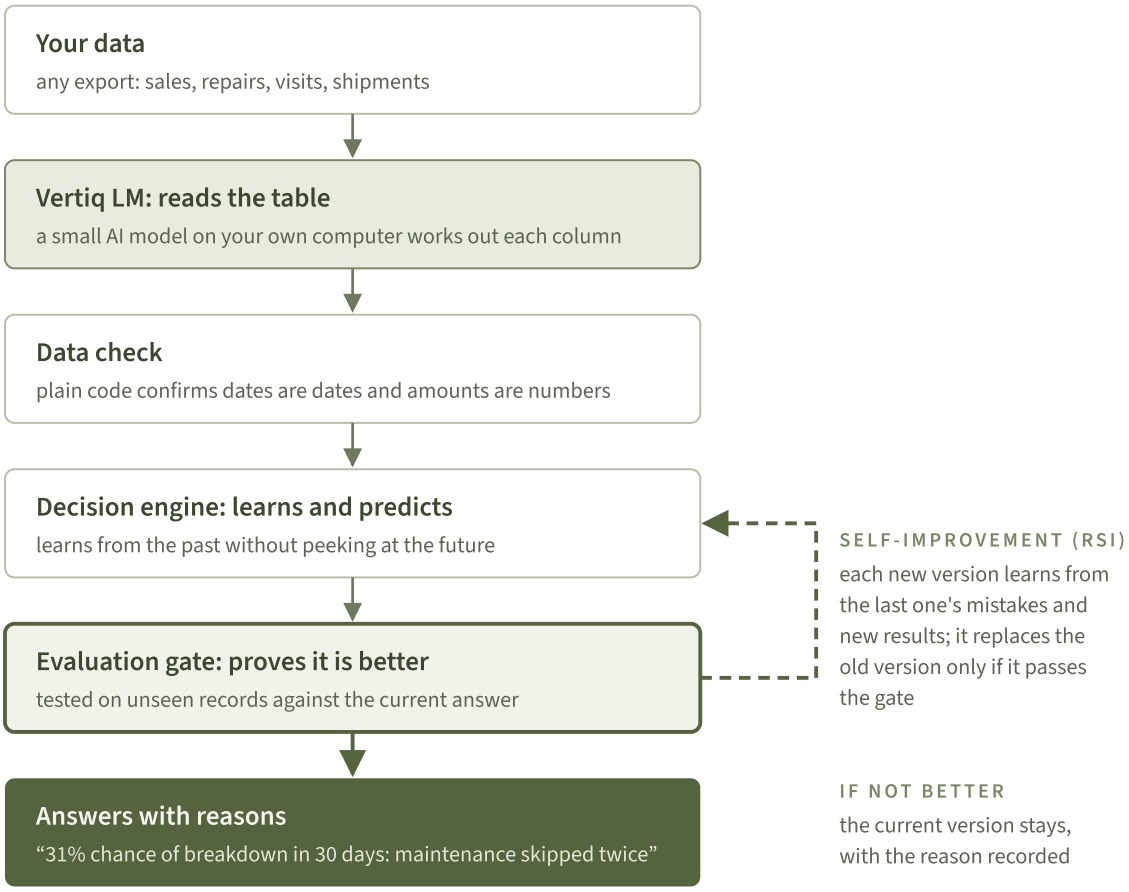
Vertiq looks at an organisation's own history and answers practical questions about what will happen next.

> “Which customers are about to stop buying?”

> “Which machines are likely to break down soon?”

> “Did our offer actually bring people back, or would they have come anyway?”

SIMPLE ARCHITECTURE



HOW IT WORKS

Six steps from a raw export to a proven answer

1

You give it your data

Any export works: a spreadsheet of sales, a list of machine repairs, patient visits. Every organisation names its columns differently: “Cust No”, “Kunden Nr”, “Client ID”.

2

It works out what each column means

Vertiq LM, a small AI model running on your own computer, reads the column names and a few rows and decides which column is the customer or machine, which is the date and which is the amount. Plain code then checks that answer against the actual rows. If the AI guessed wrong, the data corrects it.

3

It learns from the past without cheating

It places itself at earlier points in time, for example three months ago, and asks: “Using only what I knew then, could I have predicted what happened next?” It never uses information from the future, so its predictions are honest.

4

It tests itself before you rely on it

About one in five customers or machines is kept aside while it learns. It is then tested on exactly those. It must beat a simple common-sense rule and the version already in use. If it does not, the existing version stays and the reason is recorded.

5

It gives answers with reasons

“This machine has a 31% chance of breaking down in the next 30 days. The main reason: maintenance was skipped twice recently.”

6

It keeps improving itself

This is recursive self-improvement. Each new version learns from the previous version’s mistakes and from new results as they arrive. A new version replaces the old one only after passing the same test, so the system improves over time and never quietly gets worse.

EXAMPLE

A manufacturer with 1,200 machines across three plants

The maintenance team exports two years of work orders: machine number, date, type of work, cost, plant, fault code and whether scheduled maintenance was completed.

1. **Reading the data**
   Vertiq works out that “Equipment No” is the machine, “Work Order Date” is the date, “Order Type” separates breakdowns from planned maintenance, and “Cost” is the amount. The data check confirms every date and amount loads correctly.

2. **Learning**
   It learns how often each machine normally breaks down and notices that Press 7 has had two skipped maintenance visits and the same fault code three times this quarter. That is unusual for Press 7, even if it would be normal for an older machine.

3. **Adding expert knowledge**
   An engineer adds one rule of thumb: skipped maintenance in the last 60 days matters. Vertiq includes it, and the test decides whether it actually improves the predictions. In Vertiq’s own tests on data with a known answer, a rule like this raised prediction accuracy (AUC) from 0.735 to 0.767, against 0.674 for the simple rule.

4. **Testing**
   On machines it never learned from, its predictions beat the simple rule “machines that broke recently will break again”. Only then is it used.

5. **Answering**
   A ranked list for the week: “Press 7: 31% chance of breakdown in 30 days. Raises: maintenance skipped twice, repeated fault code. Lowers: recent bearing replacement.” The team inspects the top 50 first.

6. **Measuring and improving**
   The plant tries a new maintenance schedule on some machines and keeps others on the old one for comparison. Vertiq measures whether breakdowns actually fell. Next month a new version trains on the new results and replaces the old one only if it scores better.

QUESTIONS BY INDUSTRY

Advanced questions Vertiq answers

Labels show which part answers each question. PREDICT whether something happens within a window · ESTIMATE how much or how long · MEASURE whether an action caused a change · CHECK rule violations · DISCOVER statistically tested patterns · PLANNED not yet built

Manufacturing

- Which machines are likely to fail in the next 30 days, and why? Predict
- What will the next repair cost on each machine? Estimate
- Which faults keep returning after being fixed? Check
- Where are parts used beyond the bill of materials? Check
- Did the new maintenance schedule reduce breakdowns? Measure
- When will a machine reach a failure-risk threshold? Planned

Healthcare

- Which discharged patients are likely to be readmitted within 30 days? Predict
- How long is the next stay likely to be? Estimate
- Which follow-ups are overdue, and by how much? Check
- Did the discharge-call programme reduce readmissions? Measure
- Which wards show unusual readmission patterns? Discover
- Which appointments are likely to be missed? Predict

Scores support prioritisation and do not replace clinical judgement.

Logistics and supply chain

- Which shipments are likely to arrive late? Predict
- Which customers are shipping less and likely to leave? Predict
- How many units will each warehouse need next week? Estimate
- Which items will run out before the next delivery? Estimate
- Which routes or hubs have repeated delivery problems? Check
- Did changing carriers on a route reduce delays? Measure

Other industries

- **Banking:** which accounts are going dormant, and which withdrawals signal a client leaving? Predict
- **Insurance:** which policies will claim next quarter, and how large will the claim be? Predict · Estimate
- **Energy:** which meters or assets are likely to have an outage? Predict
- **Telecom:** which subscribers will stop recharging? Predict
- **Software:** which accounts will cancel, and did onboarding improve retention? Predict · Measure

A COMMON QUESTION

Why not just write a database query?

A query is very good at facts. It can tell you that Press 7 skipped maintenance twice. It cannot tell you that Press 7 has a 31% chance of breaking down in the next 30 days.

```
SELECT machine, COUNT(*) AS skipped_maintenance
FROM work_orders
WHERE type = 'maintenance' AND completed = 'no' AND date > today - 60 days
GROUP BY machine;
```

WHAT IS NEEDED	A QUERY	VERTIQ
Turning facts into a probability	Reports “skipped twice”	Learns from your history how often that actually led to a breakdown
Weighing many signals together	Filters on each signal separately	Combines age, history, plant and maintenance into one number, with each signal’s weight learned
Knowing which signals matter	You must already know the rule	Tests every signal and learns which ones predicted outcomes
Judging what is unusual	One fixed threshold for every machine	Compares each machine with its own normal rhythm
Proving it is right	No measure of accuracy	Tested on records it never learned from
Ranking thousands at once	A yes or no list	A ranked list, so the team acts on the highest risks first

> Sometimes the simple rule is good enough, and Vertiq says so. The common-sense rule is its baseline, and it is replaced only when the learned model predicts better on unseen data. When there is no real pattern, as in Vertiq’s tests on random data, it kept the simple rule in 5 out of 5 runs.

IN SUMMARY

What makes Vertiq different

Proven before it is used
A new version never replaces the current one until it has beaten it on records it has not seen.

Cause, not coincidence
Comparing with a group that was deliberately left out shows whether an action worked or would have happened anyway.

Data stays with you
Vertiq runs on your own systems. The table-reading model runs locally, and nothing is sent to an outside AI service.

Every answer is explained
Each prediction lists what raised and lowered it. Each flagged issue includes its evidence.

Improves itself, safely
Each generation learns from the last one's mistakes and from new results, and must pass the same test before it is promoted.

One system, many industries
The same steps work for customers, machines, patients, policies, meters and shipments. Only the data changes.

Evidence so far

Vertiq is tested on synthetic data where the correct answer is known in advance, repeated five times per test:

- Equipment failure: accuracy (AUC) 0.767 with one expert rule, against 0.674 for the simple rule.
- Offer effect on order value: estimated at −20.2% when the true effect was −20%.
- Random data with no pattern: no new model was promoted in any of 5 runs.
- Vertiq LM: after the data check, 99.1% of unfamiliar test tables loaded without errors.

Validation on public real-world datasets in retail, manufacturing, logistics and subscriptions is planned next.